

ARKHANDIKA BUDI WIDODOPUTRA

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• Portofolio: <https://all-about-arkha.vercel.app/>

PROFESSIONAL SUMMARY

Detail-oriented Software Developer specializing in backend architecture, native Android engineering, and fullstack web applications. Proven track record of developing interactive, algorithm-driven applications, designing efficient database schemas, and building secure REST APIs using Next.js, Kotlin, and Node.js. Passionate about writing clean, maintainable code, implementing intelligent AI algorithms (such as RAG and backtracking), and active contribution to student engineering groups.

PROJECTS

RacePace AI — F1 Race Simulation & Pit Strategy Optimizer

github.com/arkhabw/F1PitStrategyPrediction

Developer

arkhabw.github.io/F1PitStrategyPrediction/

- Engineered a paddock telemetry race strategy simulator web app using JavaScript, Chart.js, and Google Fonts.
- Developed an AI Strategy Engine using a backtracking search algorithm with branch-and-bound pruning to evaluate thousands of pit-lap and tyre compound combinations in milliseconds.
- Integrated complex race physics simulations including non-linear tyre wear ("cliffing"), fuel weight burn-off time effects (~0.06s/lap), and quadratic wet-track tyre grip scalings.

Arnai — AI-Powered Study Companion

github.com/arkhabw/Arnai

Developer

arnai-ai.vercel.app

- Built a web AI-powered study companion using Next.js, TypeScript, and Firebase.
- Integrated secure Google OAuth and dynamic vector RAG.
- Programmed automatic AI quiz generators, visual mind mapping tools, and active Spaced Repetition flashcard schedules.

LokalMarket — A Local E-commerce Android App

github.com/Alifsw21/LokalMarket

- Developed a premier native Android e-commerce platform using Kotlin and Jetpack Compose to host Indonesian traditional heritage goods.
- Implemented structured API services, a managed SQL database, and integrated local payment gateways for a frictionless checkout process.
- Optimized search queries and page loads using image compression and edge CDN routing.

EDUCATION

UPN "Veteran" Jakarta

Jakarta, Indonesia

Bachelor of Informatics (Computer Science)

2024 – Present

- Core Coursework:** Computer Science Fundamentals, Software Engineering, AI Development, Database Systems, Algorithmic Analysis.
- Activities:** Active member in multimedia development groups and student organizations, maintaining a strong academic foundation.

EXPERIENCE

KSM Multimedia

Jakarta, Indonesia

Web Developer Member

2026 – Present

- Develop and deploy full-stack web applications, managing projects from initial UI/UX wireframes to backend logic and final deployment.
- Utilize HTML, CSS, React, and Next.js to construct high-fidelity, responsive, and accessible user interfaces.
- Design and manage database schemas (MySQL, PostgreSQL) and configure server logic using Node.js/ Express to maintain secure and scalable systems.
- Collaborate with peers to conduct code reviews, ensuring adherence to coding guidelines and security best practice.

BEMF-IK UPNVJ
Human Resource Development Staff

Jakarta, Indonesia
2025 – 2026

- Supported the strategic management, evaluation, and professional development of BEMF-IK organizational members.
- Facilitated internal coordination, cross-divisional communications, and enforced organizational guidelines to maintain operational discipline.

TECHNICAL SKILLS

Programming Languages	Kotlin, Java, PHP, JavaScript, TypeScript, Python, HTML5, CSS3
Web Development	Next.js (App Router), React, Express.js, Tailwind CSS, Streamlit
Mobile Development	Android SDK, Jetpack Compose, Kotlin Coroutines, MVVM Architecture
Database & Cloud	Database Design, MySQL, PostgreSQL, Firebase Cloud Firestore, Firebase Auth
DevOps & Tooling	Git, Docker, Postman, REST API Integration, RESTful APIs, Google OAuth
AI & Optimization	Retrieval-Augmented Generation (RAG), Backtracking Search (Branch-and-Bound), Dijkstra's Shortest Path, Fuzzy Logic, Bayesian Networks

ACHIEVEMENTS

1st Place Winner | Data Science Competition POINTER 2026 By BEMF-IK UPNVJ X DSC Universitas Indonesia

- Won 1st place out of dozens of teams by developing a predictive data model to solve complex analytical problems.
- Utilized Python and advanced statistical algorithms to clean datasets, perform exploratory data analysis (EDA), and optimize model accuracy.